CloudGuard IoT — System Analysis & Design

Scope and Limitations

Status: Initial Baseline **Version:** 1.0 **Environment:** AWS Test Lab

1. Project Scope

CloudGuard IoT is a functional prototype that demonstrates coordinated monitoring and incident handling across a selected AWS cloud test environment and virtual IoT devices.

Cloud Security (In Scope)

- AWS test environment definition.
- CloudTrail & security findings ingestion.
- Dashboard cloud status views.
- Least-privilege permission guardrails.

IoT Security (In Scope)

- Virtual IoT devices via MQTT.
- AWS IoT Core identity & policies.
- Controlled anomaly generation.
- Unusual message rate detection.

Risk & Correlation

- Explainable, rule-based risk scoring.
- Defined severity bands & thresholds.
- Multi-event correlation matching.
- Incident state tracking & history.

Response & Evidence

- Automated, policy-bound containment.
- Device/resource recovery steps.
- Centralized incident timeline.
- Reproducible test scenarios.

2. Out of Scope

- Production-grade multi-account enterprise infrastructure deployments.
- Full replacement of commercial SIEM/SOAR platforms or AWS Security Hub.
- Advanced machine-learning anomaly detection or complex threat intelligence feeds.
- Offensive security testing or real-world attack execution outside the lab.
- Mandatory physical hardware integration (ESP32 is optional).

3. Key Assumptions

- The project team maintains access to an authorized AWS sandbox environment with appropriate quotas.
- Virtual IoT endpoints will serve as the baseline testing ground for telemetry and anomaly generation.
- Synthetic or controlled security events will be utilized when native production findings are unavailable.
- Automated response actions are constrained strictly to designated, allowlisted test resources.

4. Technical & System Limitations

- **Service Dependencies:** Feature availability (GuardDuty, IoT Core) relies on specific account permissions and regional support.
- **Detection Boundaries:** Behaviors cannot be identified if underlying log streams or telemetry sources are disabled.
- **Rule-Based Scoring:** Risk scores act as prioritization metrics and do not constitute absolute proof of compromise.
- **Correlation Precision:** Logic-based correlation may yield false positives or miss complex multi-vector relationships.

5. Security and Safety Guardrails

- **Isolation:** All tests execute within isolated sandbox boundaries; target systems outside the lab are strictly prohibited.
- **Credential Hygiene:** AWS access keys, private keys, and secrets will never be committed to repository code.
- **Scoped Permissions:** Response execution roles are restricted to non-production test entities.

6. Future Enhancements

- Integration of physical ESP32 hardware boards over MQTT/TLS.
- Adding machine learning models for dynamic anomaly baseline scoring.
- Role-based access controls (RBAC) and push notification channels for the operator dashboard.